DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

COURSE HANDOUT: CS301 - SOFTWARE ENGINEERING

Academic Year: 2024-2025 | Semester: V | Credits: 4

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Course Overview:
This course provides a comprehensive introduction to modern software engineering principles, methodologies, and tools. Students will learn the complete software development lifecycle, from requirements engineering and system architecture to testing, deployment, and maintenance. Special emphasis is placed on agile practices, object-oriented design patterns, software quality assurance, and automated DevOps workflows.

Course Learning Objectives:
1. Understand software development life cycles including Waterfall, Spiral, and Agile Scrum methodologies.
2. Formulate comprehensive Software Requirements Specifications (SRS) conforming to IEEE 830 standards.
3. Apply object-oriented architectural patterns and Gang-of-Four design patterns to modular system design.
4. Execute unit, integration, and system testing using automated testing frameworks and code coverage metrics.
5. Manage software projects using effort estimation techniques (COCOMO), Git version control, and CI/CD pipelines.

Course Units & Topics:
Unit 1: Software Lifecycle & Process Models (8 Hours)
- Introduction to Software Engineering & Professional Ethics
- Waterfall, Prototyping, and Spiral Lifecycle Models
- Agile Principles, Scrum Framework, Sprints, User Stories, and Kanban

Unit 2: Requirements Engineering & Modeling (9 Hours)
- Requirements Elicitation, Feasibility Analysis, and Stakeholder Needs
- IEEE 830 Standard for Software Requirements Specification (SRS)
- Unified Modeling Language (UML): Use Case, Activity, Class, Sequence Diagrams

Unit 3: Architectural Design & Patterns (10 Hours)
- System Architecture Styles: Layered, Client-Server, Microservices, Event-Driven
- Architectural Design Principles: Low Coupling, High Cohesion
- Gang-of-Four Design Patterns: Factory, Singleton, Observer, Strategy

Unit 4: Verification, Validation & Quality Assurance (10 Hours)
- Software Testing Fundamentals: Black-box vs. White-box Testing
- Unit Testing, Test-Driven Development (TDD), Test Suites
- Cyclomatic Complexity, Control Flow Graphs, Code Coverage Analysis
- Integration, System, Regression, and Acceptance Testing

Unit 5: Project Management & Contemporary Practices (8 Hours)
- Software Cost Estimation: Function Point Analysis, COCOMO Model
- Risk Management, Software Maintenance, and Re-engineering
- Continuous Integration & Deployment (CI/CD) with Git, Docker, and Automated Pipelines

Assessment Scheme:
- Continuous Internal Evaluation (CIE): 40% (2 Internal Tests, Quizzes, Lab Work)
- Semester End Examination (SEE): 60% (Comprehensive Written Examination)